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FACULTY OF COMPUTING
BCS3263
SOFTWARE QUALITY DESIGN

EDUQUEST
MANAGE ANNOUNCEMENT AND DISCUSSION

SECTION 01B

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ABSTRACT

The Manage Announcement and Discussion Module is a web-based system that can enhance the effectiveness and efficiency of communication between lecturers, students, and administrators. This module is one step toward the future in the area of managing academic announcements and discussions in a centralized way. It is created to ease the lecturer's work in sharing important updates and to provide students with a structured platform for academic discussions. As time passes, the old-school way of managing announcements through notice boards, emails or scattered messaging apps has made communication difficult and time-consuming. The reason is that announcements often get lost, discussions become fragmented and students keep increasing over time, which makes monitoring communication even more complicated. This problem has caused another issue where lecturers and administrators struggle to store, track and moderate all records of announcements and discussions effectively. Another reason the manual process of handling communication is not relevant is that lecturers are required to repeat the same task across multiple platforms, which is tiring and inefficient. Hence, this can negatively affect teaching quality and student engagement. The objective of this module is to create a system that is easy to understand and navigate while improving efficiency. This module should help reduce the time and effort required by automating repetitive communication tasks. The EduQuest system has three distinct modules which are Manage Profile, Manage Assignments and Notes and Manage Announcement and Discussion, with this module serving as the communication center that strengthens collaboration and engagement across the platform.

INTRODUCTION

The Manage Announcement and Discussion Module is an integral communication component of the EduQuest system. As educational institutions increasingly rely on digital platforms, there is a greater need for clear communication channels that support real-time updates and collaborative interactions. This module allows lecturers to convey essential information through announcements while providing students the ability to participate in threaded discussions relating to their courses. To ensure the module delivers reliable and high-quality performance, this Quality Design Report focuses on describing how the system is structured and how quality attributes guide its behavior [4]. The design representation in this document closely follows established software architecture principles and reflects the standards defined in the SQAP.

Functionality :

Functionality ensures that all intended actions such as posting announcements, editing content, replying to discussions and reading messages operate correctly and according to system requirements. It focuses on the correctness and completeness of the system features.

Performance :

Performance is concerned with the responsiveness and processing speed of the module. It ensures that announcements and discussions load quickly and remain accessible even when the system is under high user load.

Usability :

Usability ensures that the module is easy to understand, simple to navigate and visually clear for all users. This attribute is important because lecturers and students should not require technical assistance to interact with the communication tools.

These attributes guide the architecture and determine the scenarios and tactics applied to achieve high-quality behavior.

BACKGROUND OF STUDY

There are some issues and challenges faced by managing announcements and academic discussions through traditional methods such as notice boards, emails, or messaging applications [5]. There is a need for a dedicated system that allows easy and intuitive ways to manage announcements and discussions among the large number of students and lecturers in an institution. Some issues that are found when using these fragmented tools are that they are not customizable and do not provide the functions that are truly needed for academic communication. Managing announcements through emails or chats is not intuitive and often creates more problems due to human error, such as missed messages or duplicated information. When the number of students and discussions grows larger, these platforms may hit their limitations and cannot function properly anymore. Finally, collaboration is not possible in a structured way within those tools. Even though some platforms offer group messaging or shared documents, the fundamental of those systems is still not designed for academic communication and does not completely eliminate the issues concerned.

To address the issues mentioned, a web-based announcement and discussion management system is proposed. This new module aims to streamline and simplify the process of managing academic communication by providing tailor-made and scalable solutions.

The development of the EduQuest Teaching and Learning Management System involves a thorough analysis of multiple modules to provide comprehensive educational management. The modules that will be developed for the system are the “Manage Profile”, “Manage Assignment and Notes”, and “Manage Announcement and Discussion” modules, which play an important role in supporting proper administration, communication, and monitoring of student engagement.

Overall, integrating these modules into the system creates a strong foundation for handling academic communication and ensuring that information is delivered effectively and discussions are conducted in an organized and collaborative manner.

INTRODUCTION TO QUALITY ATTRIBUTES

This figure shows the Data Flow Diagram (DFD) of the module to help illustrate how information moves across the system. The DFD shows the interaction between users (lecturer, student, admin) and the internal processes such as creating announcements, managing discussions, storing records and retrieving content. Data stores such as the Announcement Database and Discussion support all CRUD activities. The DFD also highlights how user inputs flow into the system processes and how outputs are generated and returned to the interface. This structural flow ensures that quality attributes such as functionality, performance and usability can be achieved by clearly defining how data enters, moves and exits the system.

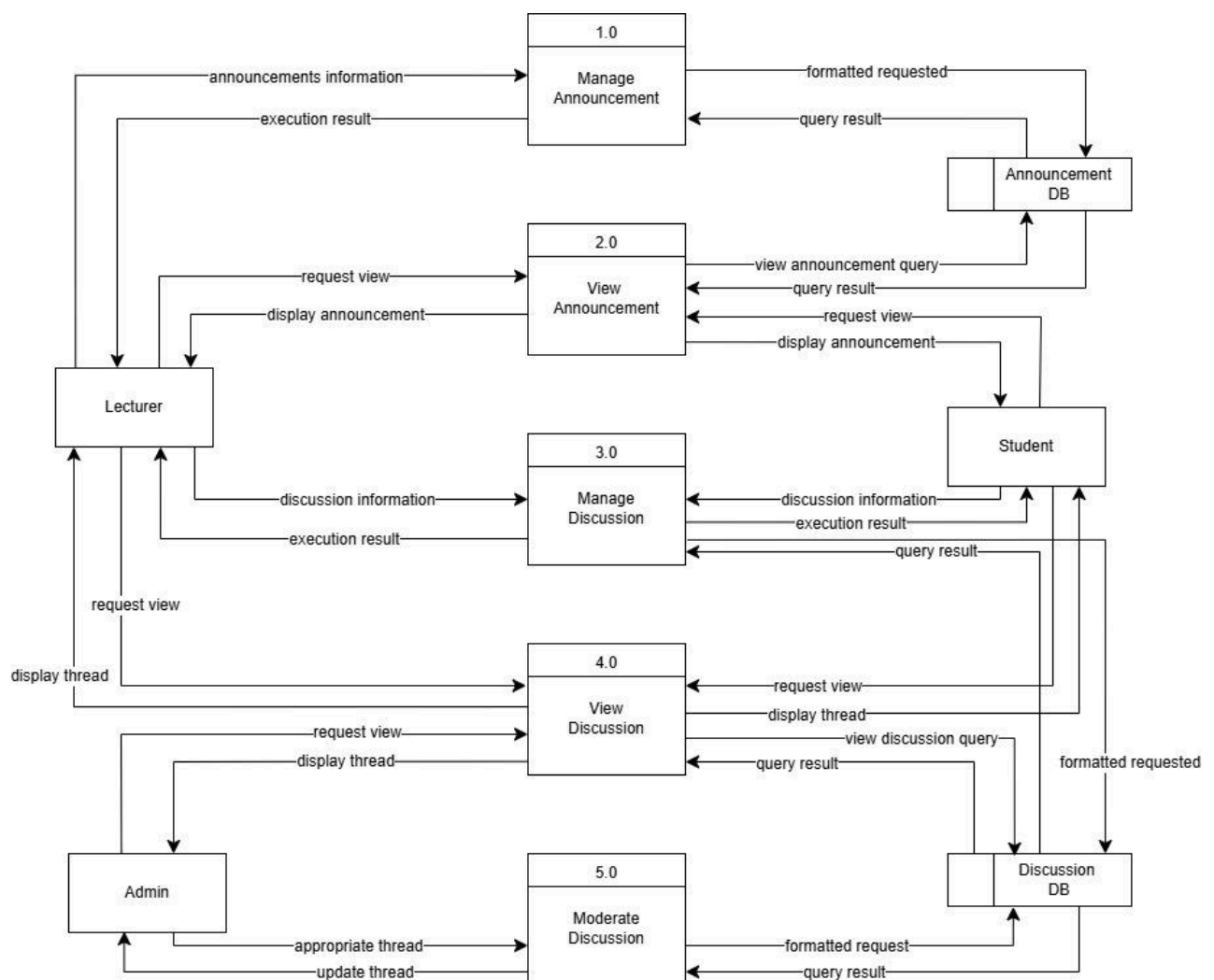


Figure 1 : DFD Manage Announcement & Discussion

There are 3 quality attributes that apply to this module. The qualities are as follows:

1. Functionality

Functionality refers to the ability of the system to perform its intended operations correctly and completely. This is a critical quality in software, as it ensures that all features behave as expected and meet the defined requirements. In this module, functionality focuses on enabling lecturers to post, edit and delete announcements, students to view and reply to discussions and administrators to moderate content. Each user role must be able to perform its designated actions without error or confusion. The system must also handle data accurately and prevent unauthorized access to sensitive operations.

- a. The module supports CRUD operations for announcements and discussion threads across all user roles.
- b. Each feature behaves according to its intended purpose and returns correct results.
- c. Role-based access ensures that only authorized users can perform specific actions.

2. Performance

Performance refers to how efficiently the system responds to user actions and handles data processing under various conditions. In this module, performance is essential to ensure that announcements and discussions load quickly even when accessed by many users at the same time. During peak academic periods, such as exam weeks or assignment deadlines, the system must remain responsive and stable. Slow loading times or delays in posting replies can disrupt communication and reduce user satisfaction. Therefore, the module must be optimized to handle concurrent usage and maintain fast response times.

- a. The module maintains fast loading and posting speeds under high user traffic.
- b. Database queries and content retrieval are optimized for efficiency.
- c. The system remains stable and responsive during peak usage periods.

3. Usability

Usability refers to how easy and intuitive the system is for users to interact with. This quality ensures that users can navigate the module, understand its features and complete tasks without requiring technical assistance. In this module, usability focuses on providing a clean and consistent interface for lecturers, students and administrators. The layout must be clear, buttons and forms must be easy to identify and the overall experience should guide users naturally through their tasks [3]. If the interface is confusing or inconsistent, users may struggle to post announcements or participate in discussions effectively.

- a. The module provides a consistent and intuitive interface for all user roles.
- b. Forms and buttons are clearly labeled and easy to interact with.
- c. Users can complete tasks such as posting, replying and moderating without confusion.

QUALITY ATTRIBUTES SCENARIO & TACTICS

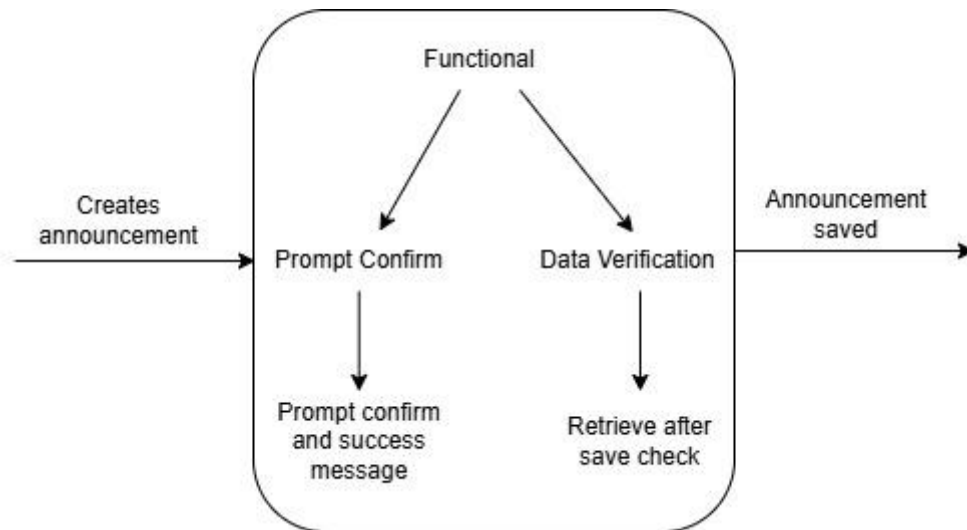
1. Functionality

Scenario 1 : Lecturer Creates Announcement

When a lecturer attempts to create a new announcement, the system provides a structured and well-organized form that allows the lecturer to enter the required details. After submission, the system verifies the input, saves the announcement accurately into the database and displays it within the announcement list. The process supports correctness and proper execution of lecturer-specific functions.

Item	Description
Source	Lecturer
Stimulus	Creating and submitting a new announcement
Artefacts	Announcement form and announcement table
Environment	Normal usage
Response	Announcement is correctly saved and displayed
Response Measure	The announcement is successfully saved and displayed in the correct list at least 95% of the time without data inconsistency or missing fields

Tactics :



Tactic Descriptions :

a. Prompt Confirmation

The system displays a confirmation dialog before final submission to ensure that the lecturer intentionally wants to proceed. This reduces accidental postings and reinforces deliberate system interaction.

b. Data Validation

The system performs strict validation checks on all input fields such as title length, content completeness and formatting to ensure the announcement meets required standards before saving.

c. Retrieve After Save Verification

After storing the announcement, the system re-queries the database to retrieve the newly saved record. This ensures that the stored information matches the lecturer's input and that no corruption or partial writing occurred.

d. Logging

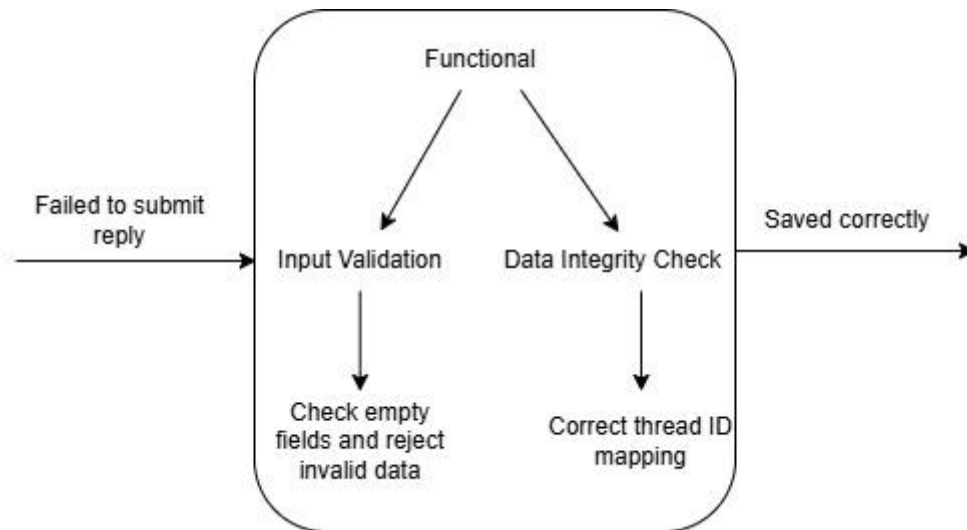
System logs record the lecturer's action, including timestamp, announcement ID, and action type. This supports traceability, debugging, and auditing for future reference

Scenario 2 : Student Posts Reply in Discussion

When a student submits a reply within a discussion thread, the system validates the reply content ensures the correct thread association and stores the reply as part of the ongoing discussion. The system consistently maintains content accuracy and thread integrity to support effective academic communication.

Item	Description
Source	Student
Stimulus	Submitting a reply
Artefacts	Reply model and thread controller
Environment	Multiple active discussions
Response	Reply is stored under the correct thread
Response Measure	Replies are correctly posted under the intended thread in at least 96% of cases without misplacement or duplication

Tactics :



Tactic Descriptions :

a. Input Validation

The system evaluates student input to ensure replies are meaningful, non-empty and properly formatted. Invalid or empty replies are rejected with immediate user feedback.

b. Thread-ID Integrity Check

Before saving the reply, the system verifies that the thread ID is valid and active. This prevents the reply from being stored under an incorrect or outdated discussion.

c. Atomic Transaction

Reply submission is wrapped in a single, indivisible database transaction. Either the entire operation succeeds or it fails completely, eliminating the risk of partial or duplicated entries.

d. Consistency Verification

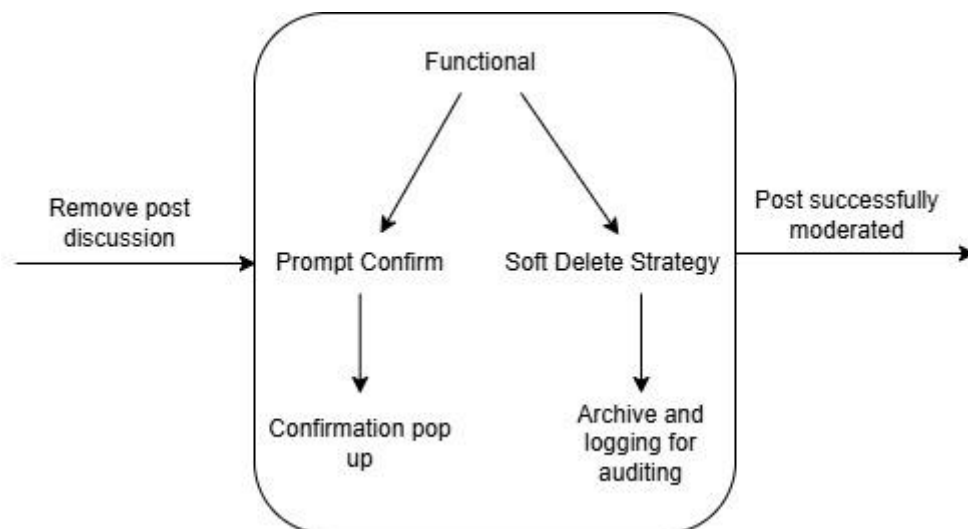
Once stored, the system reloads the reply to ensure it appears correctly within the thread and matches the submitted content.

Scenario 3 : Admin Moderates Posts

An administrator moderates discussion content by removing inappropriate posts. The system verifies the action, implements soft deletion to preserve historical data and ensures that the removal process does not disrupt discussion structure or user interactions.

Item	Description
Source	Admin
Stimulus	Deleting a post
Artefacts	Moderation tools and log table
Environment	Active discussion environment
Response	Post is successfully moderated
Response Measure	Moderated posts are successfully removed or hidden at least 97% of the time without affecting other discussion elements

Tactics :



Tactic Descriptions :

a. Confirmation Before Deletion

The system displays a confirmation dialog to ensure that the administrator intentionally wishes to remove the post, reducing accidental content removal.

b. Warning Message

Alerts admin of consequences before deleting posts with replies.

c. Soft Delete

Instead of permanently removing records, the system marks posts as hidden [2]. This ensures historical traceability and supports possible restoration for audit purposes.

d. Logging

Every moderation action is recorded with admin details, timestamp and action type to facilitate accountability and system auditing.

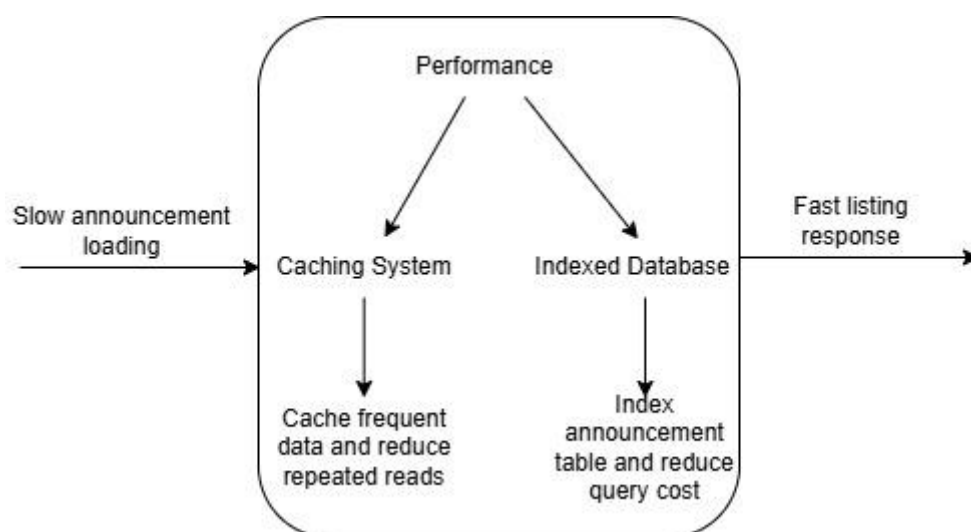
2. Performance

Scenario 1 : Viewing Announcement List Under High Traffic

During periods of high system usage, lecturers and students access the announcement list simultaneously. The system retrieves and displays announcement data efficiently, maintaining consistent performance regardless of increased user load.

Item	Description
Source	Lecturer and Student
Stimulus	Opening announcement list page
Artefacts	Announcement table and cache storage
Environment	Peak usage period
Response	Page loads quickly
Response Measure	The announcement list renders successfully without noticeable degradation in at least 94% of high-traffic interactions

Tactics :



Tactic Descriptions :

a. Caching

Frequently accessed announcements are temporarily stored in cache memory to minimize repeated database queries, improving page load stability during heavy traffic.

b. Database Indexing

Indexes on frequently queried columns enhance retrieval speed by reducing the number of records scanned resulting in faster loading and reduced system strain.

c. Remain Loading

The system prioritizes loading recent announcements first, enabling users to view immediately relevant information while background data continues loading.

d. Optimized Queries

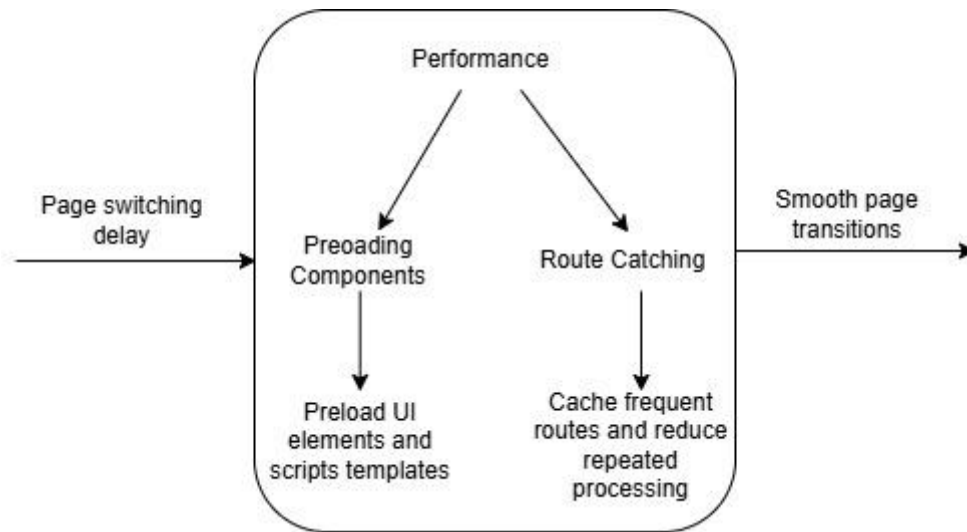
Database queries are structured to retrieve only essential fields and avoid unnecessary joins, reducing execution time and ensuring responsive performance.

Scenario 2 : Switching Between Announcement and Discussion Pages

Users frequently switch between the announcement and discussion interfaces. The system ensures that page transitions remain smooth and responsive by optimizing component loading and reducing repetitive processing.

Item	Description
Source	Lecturer and Student
Stimulus	User rapidly switches between “Announcements” and “Discussions” pages
Artefacts	Controller functions, routing system and page components
Environment	Normal usage
Response	Pages switch smoothly with minimal delay
Response Measure	Page transitions remain smooth and uninterrupted in at least 95% of navigation attempts

Tactics :



Tactic Descriptions :

a. Preloading Essential Components

The system preloads the next page's core elements (scripts, templates, small data) in the background. This reduces waiting time when the user switches pages.

b. Route Caching

Frequently accessed routes (Announcements, Discussions) are cached so the controller and view do not fully reload with every click.

c. Minimized Front-End Assets

Reducing unnecessary JavaScript, CSS and images, pages refresh faster and consume fewer resources.

d. Lightweight API Responses

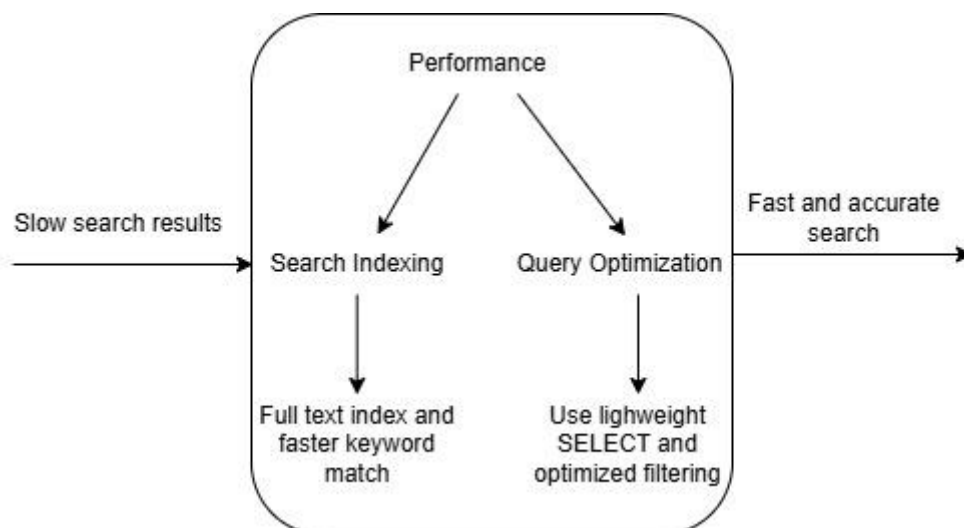
Essential data is returned during page switching (titles, previews). Full data loads later only when needed.

Scenario 3 : Searching Announcements

Users search for announcements by entering keywords into the system's search bar. The system processes the query efficiently and returns relevant results even when handling a large dataset.

Item	Description
Source	Lecturer and Student
Stimulus	Entering search keywords
Artefacts	Search function and announcement table
Environment	Large dataset
Response	Accurate results appear fast
Response Measure	Search results are returned correctly and completely in at least 96% of search operations

Tactics :



Tactic Description :**a. Full-text Search Indexing**

The system maintains an optimized index for text fields, allowing rapid keyword matching and eliminating the need for full table scans.

b. Lightweight Query

Search queries retrieve minimal required data such as title and date, reducing processing time and improving responsiveness.

c. Caching Search Results

Common or repeated search queries are cached reducing database load and accelerating future searches.

d. Optimized Filtering

The system applies filtering rules early during the search process to narrow down data efficiently and minimize unnecessary processing.

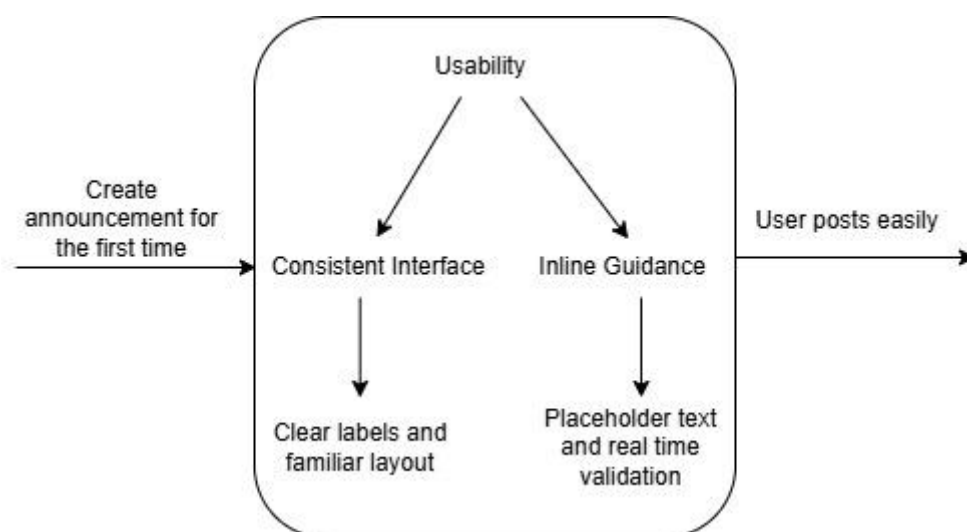
3. Usability

Scenario 1 : Lecturer Uses “Create Announcement” for the First Time

A lecturer interacting with the module for the first time navigates the announcement creation form. The interface layout, labels, hints and validation messages guide the lecturer effectively to complete the task without confusion or additional help.

Item	Description
Source	Lecturer
Stimulus	Filling in announcement form
Artefacts	Input fields and UI components
Environment	First time usage
Response	Easy completion of task
Response Measure	First-time lecturers successfully complete the form in at least 92% of attempts without needing external assistance

Tactics :



Tactic Descriptions :

a. Consistent Interface Layout

The interface follows standardized placement of buttons, labels and forms, reducing cognitive load and enabling users to develop predictable navigation patterns.

b. Clear Labels and Icons

Each component is labeled with meaningful text and recognizable icons enabling users to understand actions quickly without trial and error.

c. Placeholder Hints

Input fields include instructional placeholder text that guides users on expected formats and content types.

d. Inline Validation

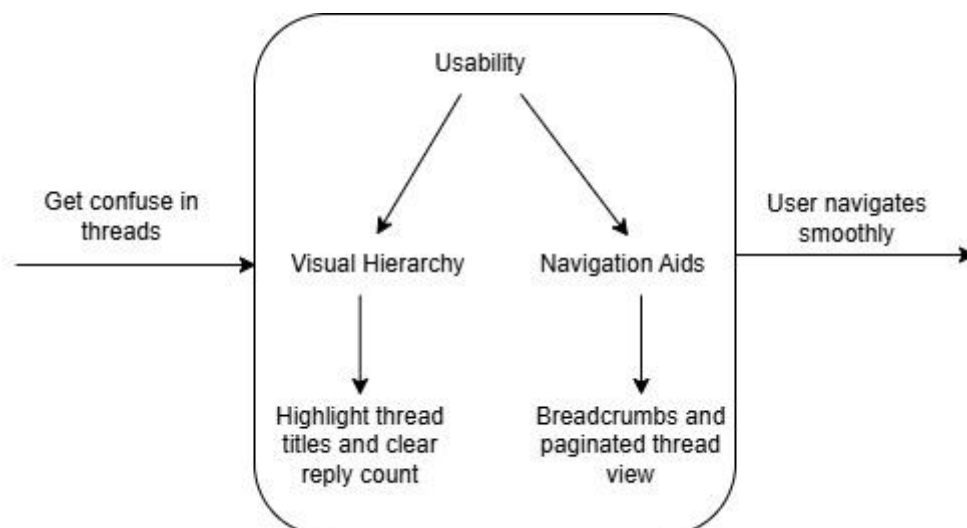
Real-time error indicators notify users of incorrect or missing input immediately preventing form submission issues and reducing frustration.

Scenario 2 : Student Navigates Discussion Threads

Students browse through discussion threads to locate relevant academic conversations. The system presents information clearly and organizes the list in a way that aids quick identification of important threads.

Item	Description
Source	Student
Stimulus	Browsing threads
Artefacts	Thread list UI
Environment	Normal usage
Response	Student finds relevant threads easily
Response Measure	Students locate their intended discussion thread in at least 93% of navigation attempts without confusion

Tactics :



Tactic Descriptions :

a. Visual Hierarchy

Key elements such as thread titles, reply counts and latest activity timestamps are visually emphasized, enabling students to identify relevant threads quickly.

b. Breadcrumb Navigation

A hierarchical navigation trail is provided, allowing students to understand their current position and move backward through discussion levels easily.

c. Pagination

The system divides lengthy discussion lists into manageable pages reducing visual overload and improving browsing clarity.

d. Simple Navigation Icons

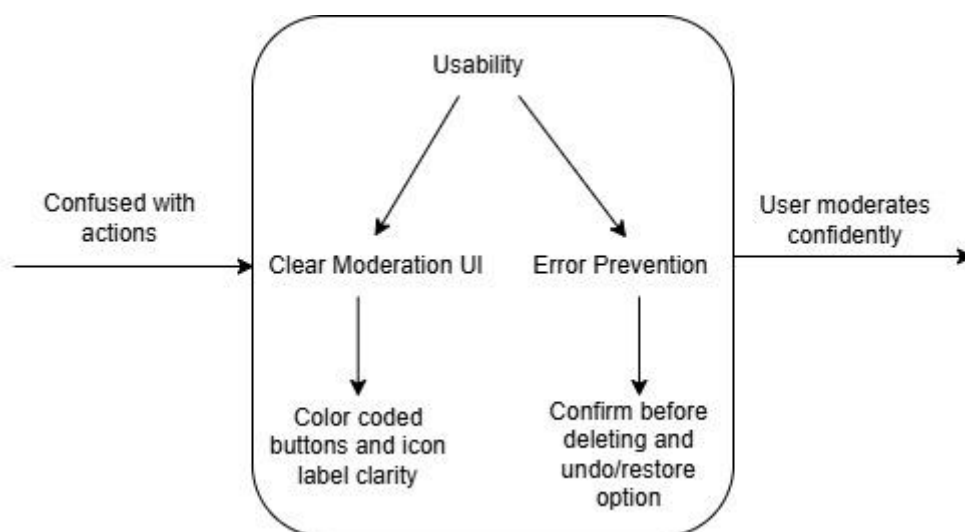
Icons for viewing and replying threads are represented using intuitive symbols for supporting quick recognition and reducing navigation effort.

Scenario 3 : Admin Moderates Content

Administrators regulate inappropriate content using moderation tools [1]. The system facilitates clear decision-making through visual cues, confirmations and recovery options, ensuring safe and accurate moderation actions.

Item	Description
Source	Admin
Stimulus	Moderating inappropriate posts
Artefacts	Moderation tools
Environment	Active discussions
Response	Admin moderates accurately
Response Measure	Administrators perform moderation actions correctly in at least 97% of attempts without making unintended deletions

Tactics :



Tactics Descriptions :

a. Color-Coded Buttons

Moderation actions are differentiated by color such as red for delete and yellow for warn which helping administrators make instant and accurate decisions.

b. Confirmation Before Delete

Before executing deletion, the system prompts the administrator with a confirmation dialog to prevent accidental removal of important content.

c. Undo or Restore Option

Soft-deleted posts can be restored, ensuring that accidental deletions can be reversed and reducing long-term data loss.

d. Clear Labels

Action buttons include descriptive text labels beneath icons, reducing misinterpretation and improving clarity during moderation.

CONCLUSION

The Manage Announcement and Discussion Module integrates key quality attributes functionality, performance and usability to deliver a reliable and efficient communication platform within the EduQuest system. Through clear scenarios and targeted tactics, the design ensures that all CRUD operations for announcements and discussions are executed correctly, securely and without data inconsistency. Performance-focused tactics such as caching, indexing and optimized queries guarantee fast loading and stable responsiveness even during high-traffic academic periods. Usability enhancements including consistent layouts, meaningful labels and intuitive navigation, support a smooth and accessible experience for all user roles.

The supporting architecture illustrated through the DFD and scenario-based analysis, provides a structured understanding of data flow and system interactions, reinforcing traceability and alignment with project requirements. Overall, this quality-driven design establishes a dependable, scalable and user-centered module that enhances academic communication and supports effective collaboration across the EduQuest platform.

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